Next-generation Bayesian local robustness

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Berkeley Neyman Seminar
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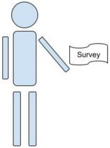
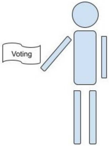
The basic problem

We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to "do you support candidate Z")

We want the average response in a target population, in which we observe only covariates.





Observe (x_i, y_i) for $i = 1, \ldots, N_S$ | Observe x_j for $j = 1, \ldots, N_T$

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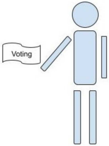

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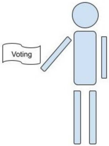

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This talk will study "Multilevel regression and poststratification" estimators:

- Set $\mathbb{E}\left[y|x,\theta\right] = m(\theta^\intercal x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\text{Survey data})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu}^{\text{MrP}}(Y_\mathcal{S}) = \frac{1}{N_T}\sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[y|x_j\right]$.

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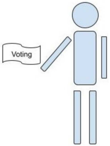

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Problem: MCMC is time consuming and the regression model is surely misspecified.
Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

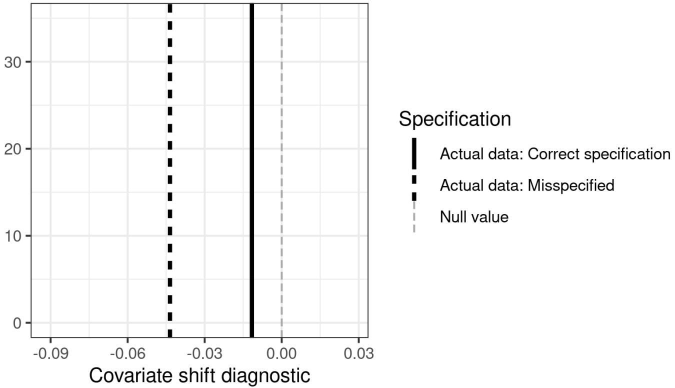

Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecied model is more extreme.

Problem: What is "large enough" to be real evidence of misspecification?
Possible answer: Frequentist confidence intervals via the parametric bootstrap.

Result preview

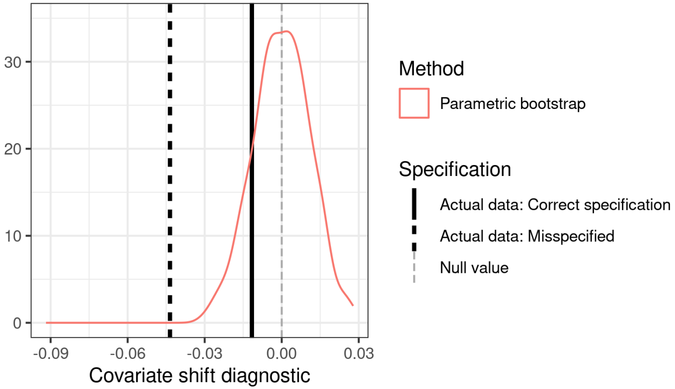

Figure 1: A covariate shift diagnostic

The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.
Possible answer: Infintesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

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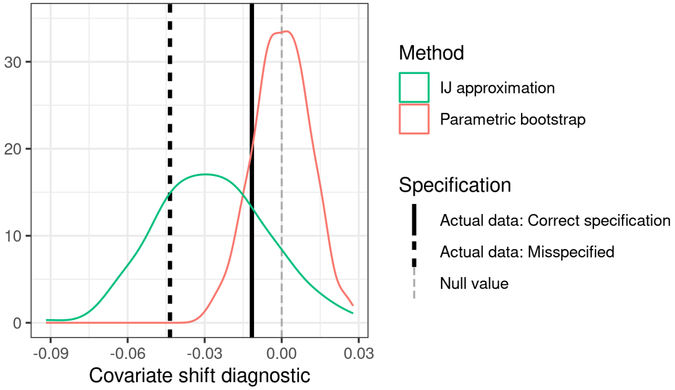

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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

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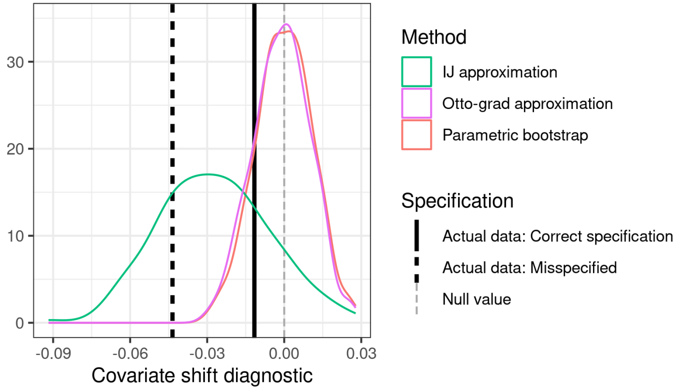

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Answer: Our flow-based local sensitivity measures ("Otto grad") produce good approximations to the bootstrap at negligible computational expense.

Outline

- Bayesian local regression specification checks
  - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
  - (With Lucas Schwengber)

Bayesian local regression specification checks

Local specification checks

$$\hat{\mu}^{\text{MrP}}(Y_{\mathcal{S}}) = \frac{1}{N_T}\sum_{j=1}^{N_T} \hat{y}_j \text{ where } \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[m(\theta^\intercal x_j)\right]}_{\text{Want to check without re–running MCMC}}.$$

This model of *survey data* is definitely misspecified.

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

Correct specification $\Rightarrow$ Regression estimates are invariant to $\mathcal{P}_S(x)$

Idea: Check invariance to $\mathcal{P}_S(x)$ directly.

Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y.
Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x)\mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x))\, \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial\mu(\mathcal{P}_\epsilon)/\partial\epsilon$
5. Plug the empirical distribution into $\Delta(w, \hat{\mathcal{P}}_S)$ and check whether ≈ 0

Bayesian posterior functionals

We are interested in $\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}[g(\theta)]$ where $g(\theta) := \frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j)$.
Let the regression log likelihood be $\ell(y|x,\theta)$.
Define the "generalized posterior" functional:

$$T(\mathcal{P}, N) := \frac{\int g(\theta) \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right)\pi(\theta)d\theta}{\int \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right)\pi(\theta)d\theta}.$$

Let $\hat{\mathcal{P}}_S$ denote the empirical distribution over x_i, y_i. Then

$$\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}[g(\theta)] = \frac{\int g(\theta)\exp\left(N\frac{1}{N}\sum_{n=1}^N \ell(y_i|x_i,\theta)\right)\pi(\theta)d\theta}{\int \exp\left(N\frac{1}{N}\sum_{n=1}^N \ell(y_i|x_i,\theta)\right)\pi(\theta)d\theta} = T(\hat{\mathcal{P}}_S, N).$$

The directional derivative is

$$\left.\frac{\partial T(\mathcal{P}_\epsilon, N)}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{P},N)}\left(g(\theta), N\int \ell(y|x,\theta)w(dx)\mathcal{P}(y|x)dxdy\right)$$

Our diagnostic is then given by

$$\Delta(w, \hat{\mathcal{P}}_S) = \mathrm{Cov}_{\mathcal{P}(\theta|\text{Survey data})}\left(\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j), \sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta)\right)$$

Simulation

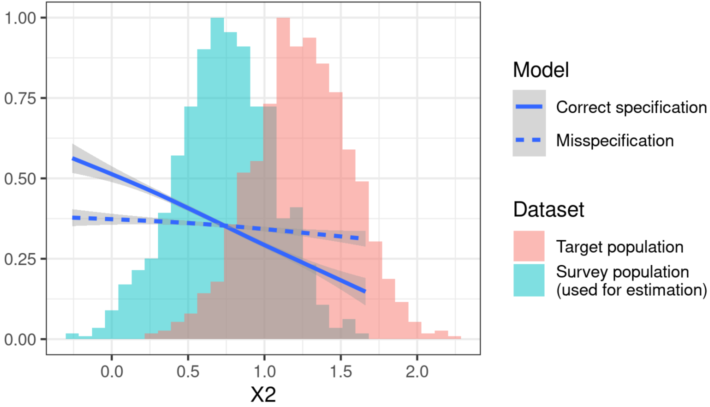

Figure 2: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

$$\text{Model 0 (correct)} : \mathtt{y} \sim \text{Logistic}(\mathtt{1 + X1 + X2 + X3 + X4 + X5})$$

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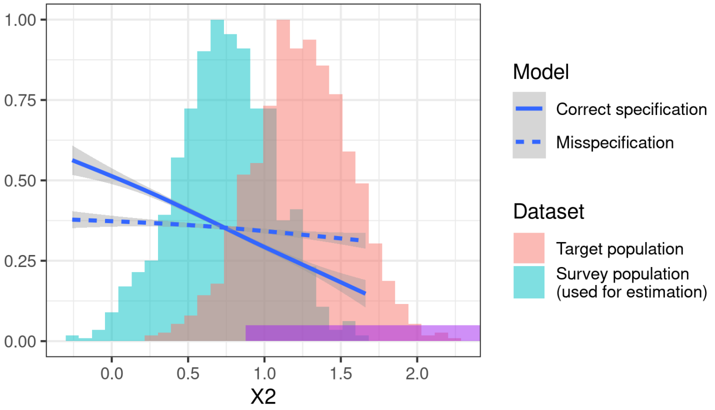

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Compute diagnostic with $w(x) = \mathbb{I}\left(\mathtt{X2} > 0.878\right)$

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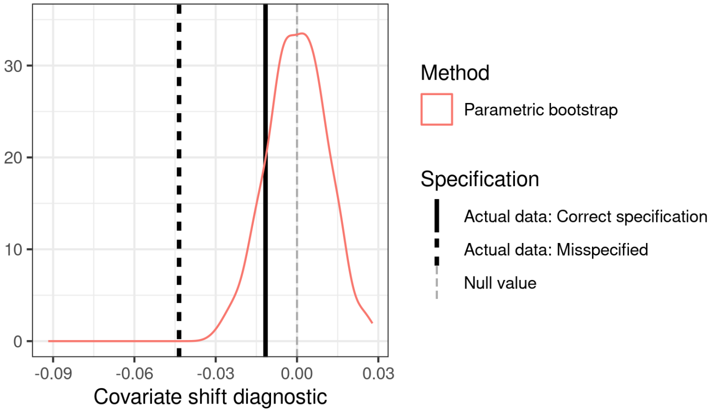

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Frequentist variance with the infinitesimal jackknife

Why frequentist variance?

For the remainder of the talk, focus on approximating the *frequentist* sampling distribution under $x_i, y_i \overset{iid}{\sim} \mathcal{P}_S(x, y)$ of quantities like our diagnostic:

$$\Delta = \mathrm{Cov}_{\mathcal{P}(\theta|\text{Survey data})}\left(\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j), \sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i, \theta)\right)$$

No need to commit to our covariate shift diagnostic to care!

- Frequentist variablity of *any* model checking diagnostic is interesting
- Frequentist variablity remains meaningful under model misspecification (Huggins and Miller, IJ papers)

We will form *local approximations* computable only from *the original posterior*.

We will compare with computationally expensive bootstrap:

- Compute Δ on the original dataset
- For $B = 1, \ldots, 100$:
    1. Sample $(x_1^*, y_1^*), \ldots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
    2. Run MCMC to estimate $\mathcal{P}(\theta|(x_1^*, y_1^*), \ldots (x_{N_S}^*, y_{N_S}^*))$
    3. Use the draws to estimate Δ_B^*
- Compare Δ with quantiles of $\Delta_B^* - \Delta$

Bayesian von–Mises Expansion

Define the "generalized posterior" functional:

$$T(\mathcal{P}, N) := \frac{\int g(\theta) \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right) \pi(\theta)d\theta}{\int \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right) \pi(\theta)d\theta}.$$

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$$\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[g(\theta)\right] = \frac{\int g(\theta) \exp\left(N \frac{1}{N}\sum_{n=1}^{N} \ell(y_i|x_i,\theta)\right) \pi(\theta)d\theta}{\int \exp\left(N \frac{1}{N}\sum_{n=1}^{N} \ell(y_i|x_i,\theta)\right) \pi(\theta)d\theta} = T(\hat{\mathcal{P}}_S, N).$$

To derive frequentist variance estimates, study the *von Mises expansion*.
Let $\mathcal{P}_t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[g(\theta)\right] - T(\mathbb{F}, N_S)\right) = \sqrt{N}\left.\frac{\partial T(\mathcal{P}_t, N_S)}{\partial t}\right|_{t=0}(\hat{\mathcal{P}}_S - \mathcal{P}_S) \quad + \mathcal{E}(\tilde{t})$$

$$= \underbrace{\sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1)}_{\text{Infinitesimal jackknife estimator}} \quad + \mathcal{E}(\tilde{t}).$$

where

$$\psi_i := N_S \mathrm{Cov}_{\mathcal{P}(\theta|\text{Survey data})}\left(g(\theta), \ell(y_i|x_i)\right)$$

Bayesian von–Mises Expansion Results

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[g(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) \quad + \mathcal{E}(\tilde{t}).$$

- Assume (sketch): A well–behaved MAP *maximum a posteriori* estimator $\hat{\theta}$ exists.
  - The dimension of θ is fixed as $N \to \infty$
  - The expected log likelihood has a strict maximum at θ_0
  - The observed log likelihood statisfies $\hat{\theta} \to \theta_0$
  - The expected log likelihood Hessian is negative definite at θ_0
- Assume (sketch): We can apply standard asymptotics.
  - The log prior and log likelihood are four times continuously differentiable
  - The prior is proper, and a technical set of prior expectations are finite
  - The log likelihood and its derivatives are dominated by a square–integrable envelope function for all θ in a neighborhood of θ_∞.

Theorem 3 [Giordano and Broderick, 2026] (sketch):

$$\sup_{\tilde{t}\in[0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{p} 0 \quad \text{in the Bayesian von–Mises expansion.}$$

Corollary: Let $\frac{1}{N_S}\sum_{i=1}^{N_S}(\psi_i - \overline{\psi})^2 \xrightarrow{p} V$. By the CLT applied to ψ_i,

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[g(\theta)\right] - T(\mathcal{P}_S, N_S)\right) \xrightarrow{d} \mathcal{N}(0, V).$$

References

R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2026.